

Assignment – 5: Data Visualization with Matplotlib

Aim: Data Visualization with Matplotlib, Data visualization on a dataset using matplotlib

Theory:

Data visualization involves presenting data graphically through charts, graphs, and plots to simplify understanding. It's crucial in data analysis, helping users pinpoint patterns, trends, relationships, and anomalies. This assignment utilizes Matplotlib and Seaborn, popular Python libraries for visualization.

Matplotlib forms the core of plotting capabilities, while Seaborn enhances it by creating more aesthetically pleasing and statistically rich visualizations with less code. Matplotlib's `pyplot` module offers a straightforward interface for various plots, including line charts, bar graphs, histograms, and scatter plots.

This assignment showcases diverse visualization types:

- **Scatter Plots** reveal relationships, such as between sepal length and sepal width in the Iris dataset.
- **Bar Plots** compare categorical values, making distinctions between groups clear.
- **Histograms** illustrate data distribution, like the spread of petal length values.
- **Box Plots** identify data distribution and potential outliers, comparing petal width across different species.
- **Pair Plots** provide a comprehensive overview of relationships across all numerical variables in the dataset.

Customization is vital for clarity. Features like titles, labels, colors, and layout adjustments (e.g., `set_title()`, `figsize`, `tight_layout()`) enhance readability and professionalism.

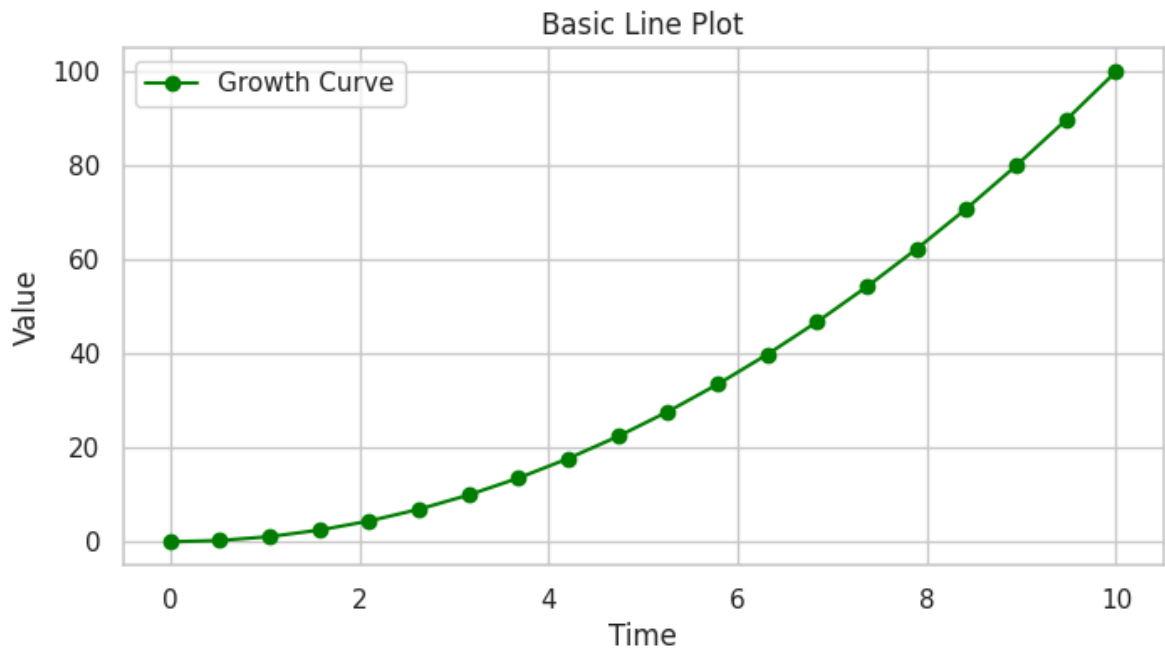
In essence, data visualization supports Exploratory Data Analysis (EDA) by simplifying complex information and fostering better comprehension before advanced analytical methods or models are applied.

```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd

x = np.linspace(0, 10, 20)
y = x ** 2
categories = ['A', 'B', 'C', 'D']
```

```
values = [5, 7, 3, 8]
```

```
plt.figure(figsize=(8, 4))
plt.plot(x, y, label='Growth Curve', color='green', marker='o')
plt.title("Basic Line Plot")
plt.xlabel("Time")
plt.ylabel("Value")
plt.legend()
plt.show()
```



```
df_iris = sns.load_dataset('iris')

sns.set_theme(style="whitegrid")

fig, axes = plt.subplots(2, 2, figsize=(12, 10))

sns.scatterplot(data=df_iris, x='sepal_length', y='sepal_width', hue='species', ax=axes[0, 0])
axes[0, 0].set_title("Scatter Plot: Sepal Dimensions")

axes[0, 1].bar(categories, values, color='skyblue')
axes[0, 1].set_title("Bar Plot: Categories")

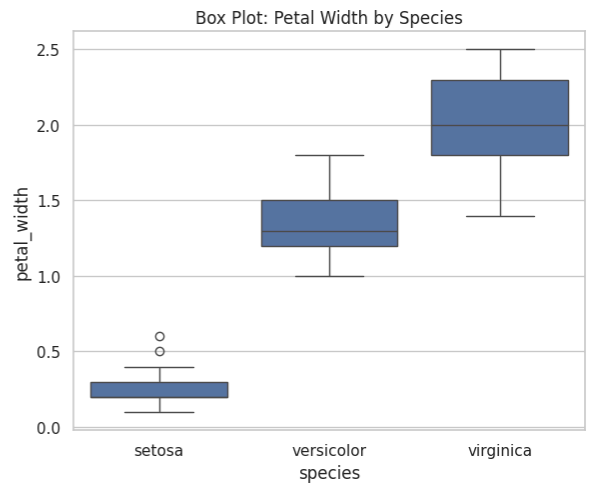
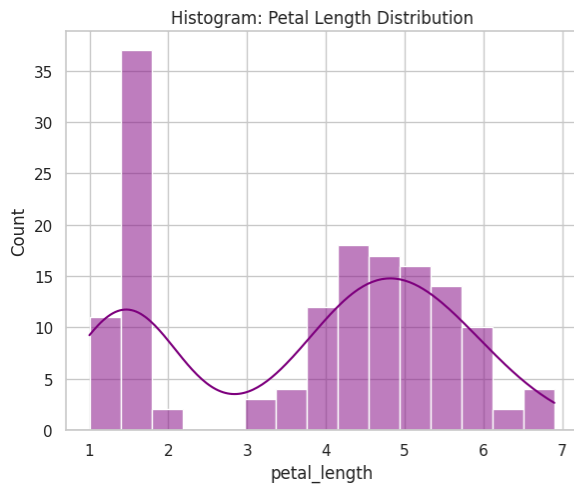
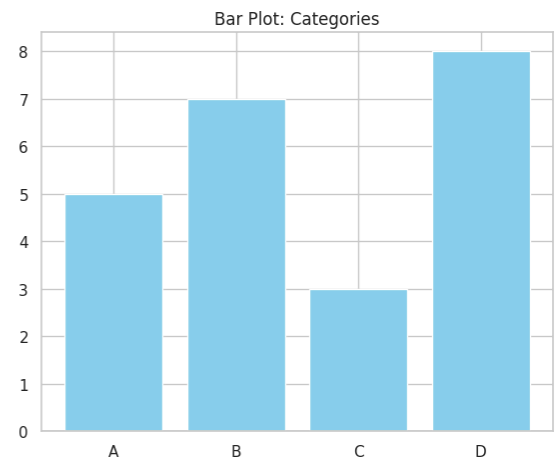
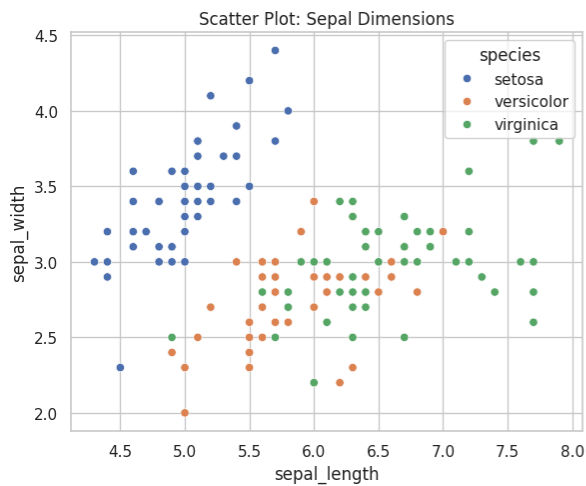
sns.histplot(df_iris['petal_length'], bins=15, kde=True, ax=axes[1, 0], color='skyblue')
axes[1, 0].set_title("Histogram: Petal Length Distribution")

sns.boxplot(data=df_iris, x='species', y='petal_width', ax=axes[1, 1])
axes[1, 1].set_title("Box Plot: Petal Width by Species")

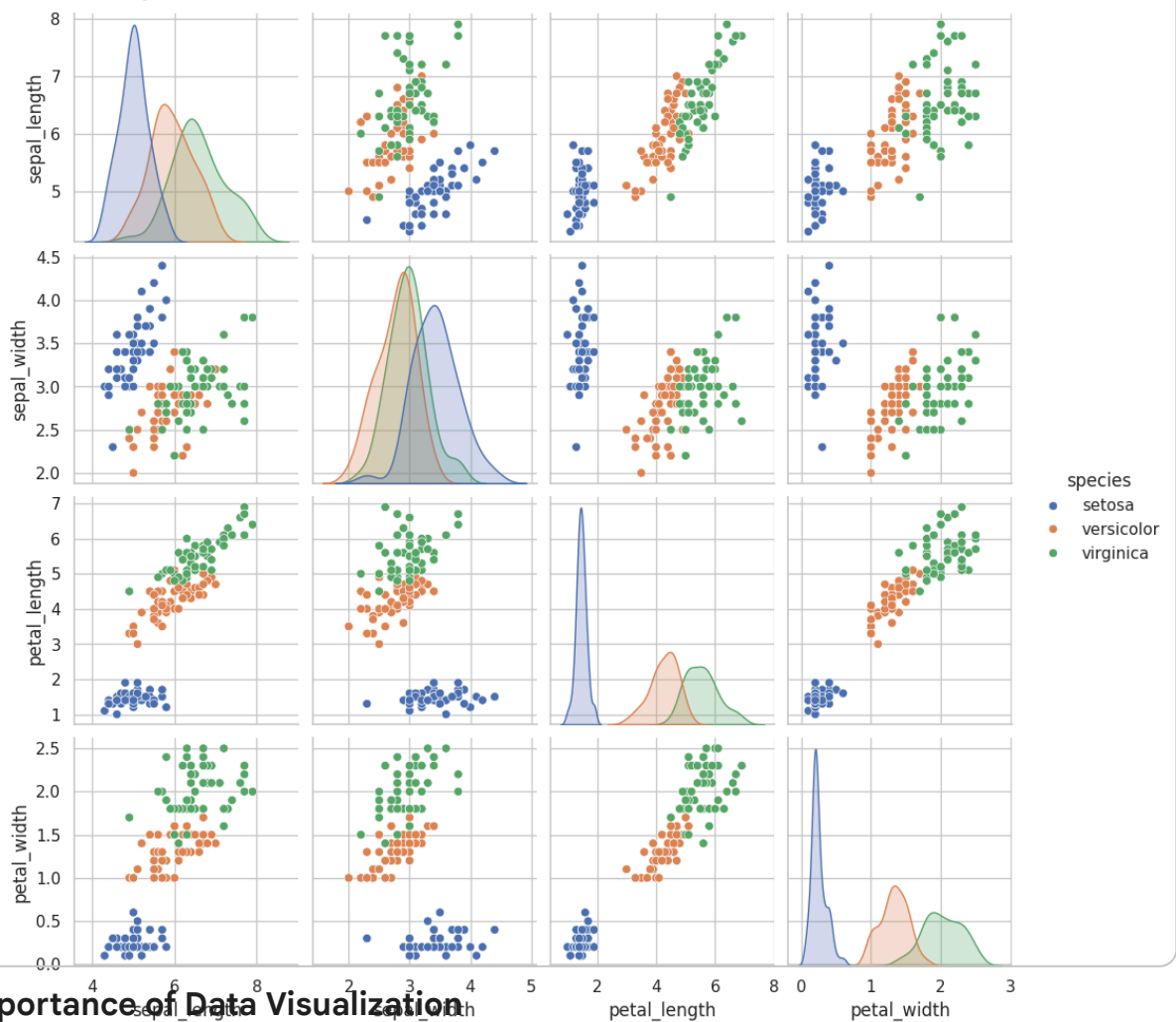
plt.tight_layout()
plt.show()

print("Generating Pair Plot...")
```

```
sns.pairplot(df_iris, hue='species')  
plt.show()
```



Generating Pair Plot...



1. Importance of Data Visualization

Data visualization is the graphical representation of data and information using charts, graphs, and plots to make data easier to understand.

1. In Analysis:

- Helps decision-makers understand complex data easily through visuals
- Makes it easier to compare data and interpret results quickly
- Supports faster and better decision-making
- Helps identify patterns, relationships, and trends in data

2. In Exploratory Data Analysis (EDA):

- Helps detect outliers and anomalies in the dataset
- Shows data distribution clearly at a glance
- Helps identify trends and patterns before applying models
- Makes it easier to understand relationships between variables

3. Purpose of Pyplot

The pyplot module in Matplotlib provides a MATLAB-like interface. It allows users to create figures, plotting areas, and decorate plots with labels without needing to manage complex object-oriented code.

4. Examples:

2. Matplotlib vs. Other Libraries

Matplotlib: The foundation of Python visualization. It offers total control over every element of a plot but requires more code for complex visuals.

Seaborn: Built on top of Matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics with less code.

Double-click (or enter) to edit

Basic Plot

Advanced Visualizations

5. Why is Customization Important?